Ch 6.2: Shrinkage - The Lasso

Lecture 18 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
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Dept of Computational Mathematics, Science & Engineering

Mon, Oct 15, 2024

Announcements

Last time:

- Ridge Regression

This time:

- The Lasso

Announcements:

- HW # 5 due Friday
- Exam next week!

Lec #	Date			Reading	HW
12	Mon	9/30	Leave one out CV	5.1.1, 5.1.2	
13	Wed	10/2	k-fold CV	5.1.3	
14	Fri	10/4	More k-fold CV,	5.1.4-5	
15	Mon	10/7	k-fold CV for classification	5.1.5	
16	Wed	10/9	Subset selection	6.1	HW #4 Due Weds 10/9
17	Fri	10/11	Shrinkage: Ridge	6.2.1	
18	Mon	10/14	Shrinkage: Lasso	6.2.2	
19	Wed	10/16	Dimension Reduction	6.3	
20	Fri	10/18	Overflow, Possibly more dimension reduction?		HW #5 Due Fri 10/18
	Mon	10/21	No class - Fall break		
	Wed	10/23	**Review**		
	Fri	10/25	**Midterm #2**		

For next year, add some slides about why we would use lasso vs ridge regression. Ridge has closed form solution while lasso requires gradient descent (confirm this) could be an option.

Section 1

Last time

Goal

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso

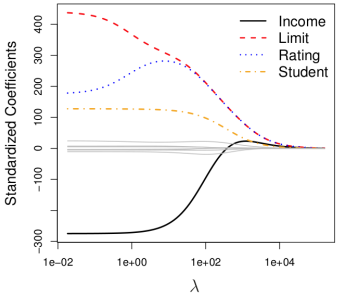
Ridge regression

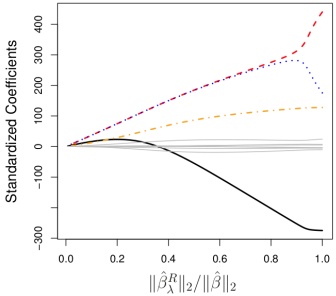
Before:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

After:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$





Scale equivariance (or lack thereof)

Scale equivariant: Multiplying a variable by c (cX_i) just returns a coefficient multiplied by $1/c$ ($1/c\beta_i$)

- Least squares is scale equivariant
- Ridge regression is not

Solution: standardize predictors

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

- Ex: `income variable`. If you put in income by \$1,000's of dollars, the answer you would get is quite different. Might emphasize a different coefficient.
- Least squares is scale equivariant
- Ridge regression very much is not
- $X_j\hat{\beta}^R_{j,\lambda}$ depends not only on λ but also on values of other predictors

Section 2

The Lasso

Same goal as before

- Fit model using all p predictors
- Aim to constrain (regularize) coefficient estimates
- Shrink the coefficient estimates towards 0 *Ridge shrunk but DID NOT GET RID OF which is what we actually want*

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

- Ridge regression
- Lasso: Least Absolute Shrinkage and Selection Operator

The lasso

Least Squares:

$$RSS = \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

Ridge:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = RSS + \lambda\sum_{j=1}^{p}\beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

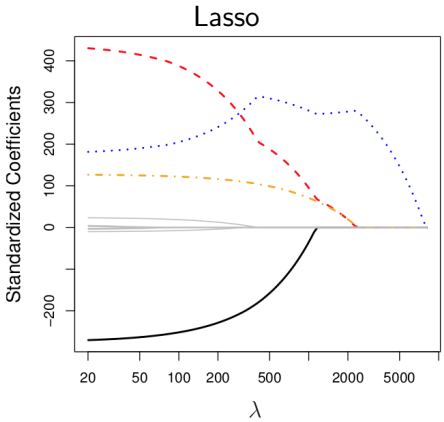
- ℓ_1 penalty instead of ℓ_2 penalty
- ℓ_1 norm: $\|\beta\|_1 = \sum|\beta_j|$
- ℓ_2 norm: $\|\beta\|_2 = \sum\beta_j^2$

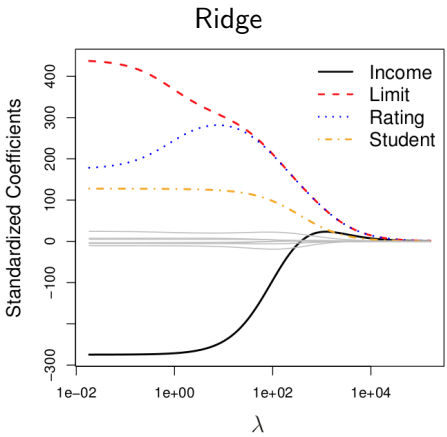
Subsets with lasso

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = RSS + \lambda\sum_{j=1}^{p}|\beta_j|$$

- Result of ℓ_1 penalty is that it forces some coefficients to be exactly 0
- Can be interpreted as something more like subset selection, but here called *variable selection*
- Yield are *sparse* models, that is those using only a subset of variables
- Selecing good λ is critical

An example on `Credit` data set





- $\lambda = 0$: Both are least squares
- $\lambda = \infty$: Everything is 0
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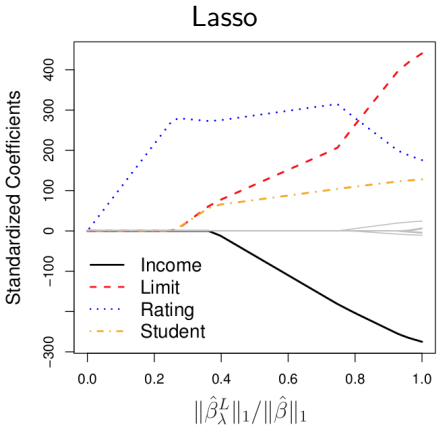
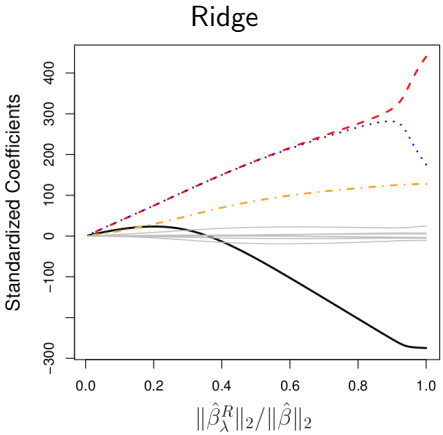
More example on `Credit` data set





- Note that ridge always includes all variables, even if small coefficient
- Lasso produces different sized subsets depending

Scale equivavariance (or lack thereof)

Scale equivariant: Multiplying a variable by c just returns a coefficient multiplied by $1/c$

Least squares **is** scale equivariant. Ridge/Lasso **are very much not**.

Solution: standardize predictors

$$\widetilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

- Ex: `income variable`.
- Least squares is scale equivariant
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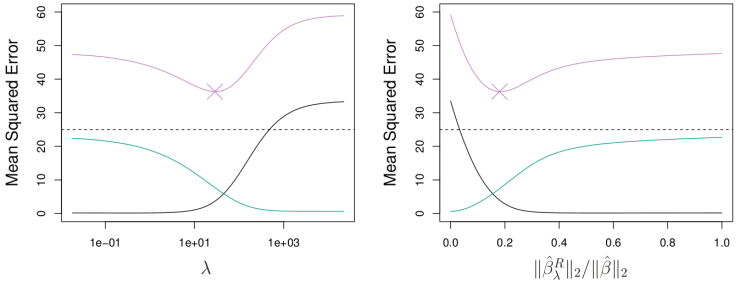
Bias-Variance tradeoff



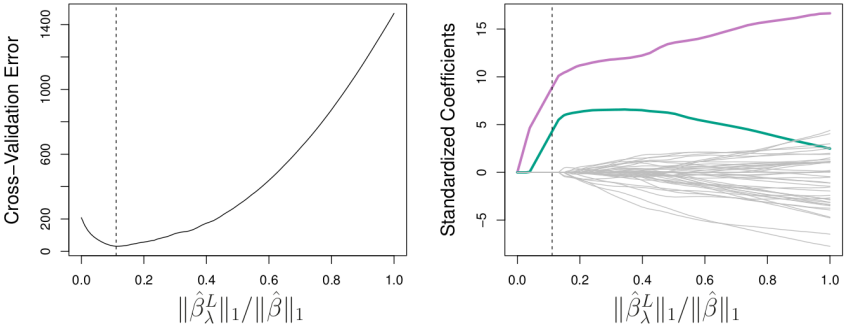
Squared bias (black), variance (green), and test mean squared error (purple) for simulated data.

Horizontal dashed line is minimum possible test MSE

Using Cross-Validation to find λ

- Choose a grid of λ values
- Compute the (k-fold) cross-validation error for each value of λ
- Select the tuning parameter value λ for which the CV error is smallest.
- The model is re-fit using all of the available observations and the selected value of the tuning parameter.

10-fold CV choice of λ for lasso and simulated data



- Data generated with $p = 45$ variables, $n = 50$ data points, only 2 variables actually involved
- The vertical dashed line: choice of λ
- Right: Two colored lines are predictors related to the response (signal), grey are unrelated (noise)
- Correctly gives much larger coefficient estimates to the two signal predictors, but also the min CV error corresponds to a set of coefficient estimates for which only the signal variables are non-zero.
- Note that least squares solution doesn't even emphasize one of the variables (green one)

Coding example

Ridge vs Lasso

Ridge Regression:

- Coefficients are shrunk towards 0, but not forced to be exactly 0
- Tends to be better than lasso when many predictors have significant effect on output
- More robust to outliers

Lasso:

- Coefficients are shrunk towards 0, and some are forced to be exactly 0
- Tends to be more interpretable since we can see which variables are non-zero (rather than having to decide how small is small enough)
- More sensitive to outliers

TL;DR - Original forumlation

Least Squares:

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

Ridge:

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = RSS + \lambda \sum_{j=1}^{p} \beta_j^2$$

The Lasso:

$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| = RSS + \lambda \sum_{j=1}^{p} |\beta_j|$$

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